Huawei Access Node WFD-ACC-18 Optical Transport Failure Resolution SOP (ALARM_1887)

Node: WFD-ACC-18 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1887

Description

This Standard Operating Procedure (SOP) is a comprehensive guide to troubleshooting and resolving Optical Transport Failures (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Huawei Access Node WFD-ACC-18 at the Whitefield POP. The Huawei Access Node WFD-ACC-18 is a critical component of the Optical Transport Network (OTN) infrastructure, responsible for transmitting and receiving high-speed optical signals. The failure of this node can have significant impacts on network availability, reliability, and overall performance. The root cause of this failure can be attributed to various factors, including SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. These issues can arise due to factors such as aging components, improper installation, or environmental stressors like temperature and humidity fluctuations. The Huawei Access Node WFD-ACC-18 employs the Optical Transport Network (OTN) protocol, which enables the efficient transmission of high-speed optical signals over long distances. The OTN protocol uses a combination of wavelength division multiplexing (WDM) and time division multiplexing (TDM) to enable the simultaneous transmission of multiple signals over a single fiber. However, the OTN protocol is not immune to the effects of SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, which can disrupt the transmission of optical signals and ultimately lead to network downtime. To mitigate these risks, it is essential to implement regular maintenance and testing procedures, including SFP/XFP inspection and cleaning, fiber inspection, and Tx/Rx power level monitoring. Additionally, network administrators must be trained to respond quickly and effectively to Optical Transport Failures, utilizing the diagnostic and resolution procedures outlined in this SOP. By following the steps outlined in this SOP, network administrators can quickly identify the root cause of the failure, implement effective resolution strategies, and restore network availability and performance.

Pre-requisite Checks

- check: Verify that the Whitefield POP is accessible and that dispatch of remote hands has been initiated, status: Completed

- check: Verify that the redundant link status is available and functioning correctly, status: Completed

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display device transceiver info, Expected Output: Transceiver Information: SFP/XFP Type: 10G-SFP+ 850nm Serial Number: WFD-ACC-18-1234 Status: Online, Description: This command displays the SFP/XFP information, including the type, serial number, and status. The expected output indicates that the SFP/XFP is online and functioning correctly.

Command: display interface gigabitethernet 0/1/0, Expected Output: Interface GigabitEthernet 0/1/0 Status: Up Rx Power: -20.5 dBm Tx Power: -15.2 dBm, Description: This command displays the interface status, including the Rx and Tx power levels. The expected output indicates that the interface is up and functioning correctly.
```

Resolution Steps

Step 1: Step: Swap the SFP/XFP module, Command: swap device transceiver 0/1/0, Description: This step involves swapping the SFP/XFP module to determine if the failure is related to the SFP/XFP or the fiber.

Step 2: Step: Inspect and clean the SFP/XFP module, Command: clean device transceiver 0/1/0, Description: This step involves inspecting and cleaning the SFP/XFP module to ensure that it is functioning correctly.

Step 3: Step: Verify the fiber connection, Command: display interface gigabitethernet 0/1/0, Description: This step involves verifying the fiber connection to ensure that it is secure and functioning correctly.

Post-Check Verification

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Command: display device transceiver info, Expected Output: Transceiver Information: SFP/XFP Type: 10G-SFP+ 850nm Serial Number: WFD-ACC-18-1234 Status: Online, Description: This command is used to verify the SFP/XFP information, ensuring that it is online and functioning correctly.

Command: display interface gigabitethernet 0/1/0, Expected Output: Interface GigabitEthernet 0/1/0 Status: Up Rx Power: -20.5 dBm Tx Power: -15.2 dBm, Description: This command is used to verify the interface status, including the Rx and Tx power levels, ensuring that it is up and functioning correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: undo swap device transceiver 0/1/0, Description: This step involves restoring the original SFP/XFP module to its original position, ensuring that the network is restored to its original state.